Preclass Learning: Web Scraping (ICA 15)

1. The Scraping Process

Web scraping is the automated extraction of data from websites using a programmatic approach.

1. **Request HTML:** Use an HTTP request function to download the raw HTML source code.
2. **Parse Document:** Convert the raw HTML string into a navigable tree structure (DOM).
3. **Locate Data:** Identify the specific HTML elements containing the desired data.
4. **Extract:** Pull the text or attribute data from those elements.
5. **Wrangle:** Clean and structure the extracted data into a tidy format.

2. Element Selection

To target specific data points, you must master selection techniques:

- **CSS Selectors:** Select elements based on tags (p), unique ID (#id), or class (.class-name).
- **XPath (XML Path Language):** A powerful language for navigating the hierarchical structure of the HTML document to target complex nested elements (e.g., the third row of the second table).

3. Ethical and Technical Boundaries

- **robots.txt:** A standard file on a domain that tells automated web crawlers which parts of the site they are *disallowed* from accessing. Always check this file first.
- **Rate Limiting:** Implement pauses between requests (using Sys.sleep()) to avoid placing too much load on the target server, which can lead to your IP being blocked.
- **JavaScript-Rendered Content:** Simple scraping tools cannot execute JavaScript. If data is loaded dynamically after the page loads (e.g., via AJAX), specialized tools like headless browsers may be required.